ITA06020 MACHINE LEARNING

Experiment 6 – Implementation of Naive Bayes Algorithm

Aim:

To implement the Naive Bayes classification algorithm in Python and display the results using a confusion matrix and accuracy.

Dataset: Breast Cancer Wisconsin Dataset (real-world medical benchmark dataset available in Scikit-learn).

Procedure

1. Import the required Python libraries.
2. Load the Breast Cancer dataset.
3. Separate the input features and target values.
4. Split the dataset into training and testing data.
5. Create a Gaussian Naive Bayes classifier.
6. Train the model using training data.
7. Predict the output using testing data.
8. Display the confusion matrix and accuracy.

Python Program

```
from sklearn.datasets import load_breast_cancer

from sklearn.model_selection import train_test_split

from sklearn.naive_bayes import GaussianNB

from sklearn.metrics import accuracy_score, confusion_matrix

data = load_breast_cancer()

X = data.data

y = data.target

X_train, X_test, y_train, y_test = train_test_split(

   X, y, test_size=0.2, random_state=42

)

model = GaussianNB()

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Naive Bayes Classification")

print("Predicted Output:", data.target_names[y_pred[0]])
```

```
print("Actual Output:", data.target_names[y_test[0]])

cm = confusion_matrix(y_test, y_pred)

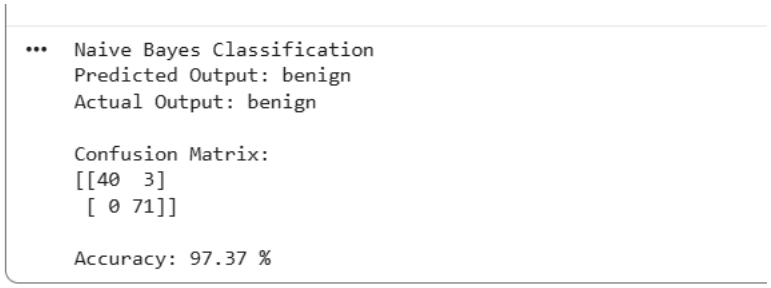
print("\nConfusion Matrix:")

print(cm)

accuracy = accuracy_score(y_test, y_pred)

print("\nAccuracy:", round(accuracy * 100, 2), "%")
```

Output:



Result

Thus, the Naive Bayes classification algorithm was successfully implemented using Python. The model achieved an accuracy of 97.37%, and its performance was evaluated using a confusion matrix.